KEVIN WILLIAMS

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SUMMARY

AI/ML engineer with four years of professional engineering experience and an M.S. in Artificial Intelligence, building LLM and RAG-enabled applications and deploying computer-vision models to production across NASA human-spaceflight and automotive manufacturing programs. Pairs applied AI development with systems-engineering and AI-assurance depth — requirements, verification, and audit of AI decision pipelines for safety-critical systems.

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow, Hugging Face (Transformers, Spaces); LLM application development retrieval-augmented generation (RAG), chat-based CRUD, structured-output generation, LangChain; sequence models (RNN / LSTM / Transformer); CLIP, FAISS vector search & embeddings; CNNs, ensemble methods; differential privacy (Opacus), adversarial robustness, membership-inference red-teaming

ML Engineering & Cloud: AWS, Docker, FastAPI, REST APIs, Git, Hugging Face Spaces deployment

Systems Engineering & AI Assurance: Requirements development & V&V, MBSE, NPR 7123 / 7150.2 / 8739, AI requirements & verification frameworks, audit / traceability, milestone reviews

Software: Python, SQL, JavaScript, Django, Flutter, MySQL

EXPERIENCE

Systems Engineer / Software Architect — Barrios Technology Jan 2026 – Present

NASA Johnson Space Center · Houston, TX

- Built a document-quality-assurance (DQA) prototype for Lunai, JSC's internal AI agent — a 4-bit-quantized Llama-3-8B few-shot classifier that screens NASA documents for export-control (ITAR) sensitivity, with automated Excel/Word reporting via a PDF-ingestion pipeline.
- Validated at 100% on a 20-example held-out set with zero false positives across sampled chunks of the public NPR 7150.2D; authored a roadmap to a fine-tuned multi-task DQA model spanning technical-accuracy, formatting-compliance, metadata, and records-retention checks.
- Designing and building the EHP Program Hub Dashboard, cataloging 100+ tools across 9 offices to consolidate tool visibility for the Extravehicular Activity and Human Surface Mobility Program; delivered as a vanilla HTML/CSS/JS single-file application under deliberate audit-friendly deployment constraints, partnering with IT and engineering leads to define schema and drive structured data collection.

Software Engineer — Barrios Technology Feb 2025 – Dec 2025

Houston, TX

- Architected and built Cortex, an enterprise data-management platform (Django + EAV + MySQL) replacing legacy SharePoint workflows for Intuitive Machines; embedded a retrieval-augmented (RAG) AI chat assistant with structured-output suggestions over program data, supporting both cloud and local model inference.
- Implemented role-based access control, dynamic user-configurable tables, and a safe formula/calculation engine; deployed internally at Barrios for dev/test validation and delivered to customer-readiness stage.

Systems Engineer — Barrios Technology Oct 2023 – Feb 2025

NASA Johnson Space Center · Houston, TX

- **Commercial LEO Development Program (CLDP):** Served as NASA SE&I reviewer across four commercial-station partners, conducting ~40 reviews in 9 months including CDPO-delegated lead review authority for Blue Origin; generated RFAs/RIDs reported to the Program Office and drove partner design changes that tightened technical baselines.
- Led requirements traceability for CLDP, tracing 300+ requirements to CLDP-REQ-1130 across three commercial providers and delivering a unified baseline adopted by the Program Office, NASA, and all partners.
- Developed an **AI requirements & verification framework** for commercial human spaceflight — authored 70 requirements building consensus across SMEs from University of Michigan, FAA, ISO, and NASA, and cofounded an AI-in-Flight working group (published via Safety Critical Labs; Zenodo DOI below).
- **Orion MPCV (Artemis I → II):** Evaluated Flight Test Objectives across 18 subsystems (aero, TPS, propulsion, GN&C, ECLS, avionics, software) to deliver the Artemis I Flight Analysis Report; integrated Lockheed Martin and NASA subsystem inputs.

- **ISS / Axiom:** Traced SSP 51074 CERD requirements for Axiom docking integration through NPR 7123 milestone reviews (SRR).

Engineering Specialist — Ford Motor Company Jun 2022 – Sep 2023
Van Dyke Electric Powertrain Center · Van Dyke, MI

- Built a deep-learning computer-vision pipeline (PyTorch regression with Kalman filtering for segmentation) for varnish surface-area measurement on eMotor stators, replacing a manual operator workflow and achieving 99% accuracy with 1+ hour saved per quality check. Recognized with the Ford Technical Excellence Award (2023).
- Led launch and served as deputy manager for a 20-person eMotor materials laboratory on 24/7 coverage for F-150 Lightning and Maverick Hybrid programs; achieved ISO certification and production readiness.

Undergraduate Research Assistant — University of Michigan Sep 2019 – Jun 2022
Advisor: Prof. M. Meyerhoff · Ann Arbor, MI

- First-author electrochemical Δ9-THC detection method on disposable screen-printed carbon electrodes (0.13 μM LOD, $R^2 = 0.995$); published in Electroanalysis (2023).

Early Career — Materials & Chemical Engineering 2012 – 2017
General Motors · Hyundai-Kia · Quaker Chemical

- Metallurgical lab buildup, material validation, and supplier quality across automotive powertrain programs.

RESEARCH & AI PROJECTS — M.S. ARTIFICIAL INTELLIGENCE

μStruct — AI-Powered Microstructure Identification (ASTM)

- Built an ensemble classifier combining CLIP zero-shot inference with FAISS similarity-weighted voting (40/60) across 6,000+ micrographs spanning 16 steel microstructural phases; productionized as a FastAPI + Docker service on Hugging Face Spaces with a REST API. Live demo: kevwill-microstructure-classifier.hf.space

Privacy-Preserving Cognitive Twin — Membership-Inference Attacks & Defenses (CIS 545)

- Designed an adversarial evaluation pipeline (confidence-based, label-only, and shadow-model attacks) and benchmarked DP-SGD (Opacus), confidence-exclusion, and a fusion defense with ε-sweeps mapping privacy–utility tradeoffs across attack success rate, AUROC, accuracy, and calibration.

Disaster Infrastructure Damage Recognition (CIS 583)

- Developed a CNN binary classifier for post-event infrastructure damage assessment and established classification confidence thresholds for first-responder deployment readiness.

EDUCATION

M.S. Artificial Intelligence — University of Michigan | GPA 3.8/4.0 Aug 2026
B.S. Mechanical Engineering — University of Michigan Dec 2021

PUBLICATIONS

- Williams, K. Requirements and Verification Standards for Artificial Intelligence in Safety-Critical Applications. Safety Critical Labs. Zenodo, 2026. DOI: 10.5281/zenodo.19323435
- Williams, K. J., White, C. J., Hunt, A. P., & Meyerhoff, M. E. Differential pulsed voltammetry of Δ9-THC on disposable screen-printed carbon electrodes. Electroanalysis, 2023. DOI: 10.1002/elan.202200451

AWARDS

- Silver Bear Award (HSFTIC, 2025) — leading role integrating AI into NASA’s Software Engineering Handbook.
- Ford Technical Excellence Award (2023) — deep-learning computer vision for eMotor stator inspection.